

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Summer Examination – 2024**

**Course: B. Tech.      Branch: Mechanical Engg./Mechanical Engg. (Sandwich)      Sem: VII**

**Subject Code & Name: BTMC701      Mechatronics**

**Marks: 60      Date: 01/07/2024**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                            | (Level/CO) | Marks     |
|------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q.1 Solve Any Two of the following.</b>                                                                 |            | <b>12</b> |
| A) What is the difference between a sensor and transducer? Explain it with specific examples.              | (CO1)      | 6         |
| B) How a capacitive level sensor works? Give its applications.                                             | (CO1)      | 6         |
| C) Discuss the working of DC Motor with a neat sketch and state its advantages.                            | (CO3)      | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                                                                 |            | <b>12</b> |
| A) Describe the Data Acquisition System with a block diagram. List any two practical examples.             | (CO2)      | 6         |
| B) Draw a neat diagram of a Seven segment display system? Explain it with an example.                      | (CO2)      | 6         |
| C) Explain with a neat sketch On/Off Controller.                                                           | (CO6)      | 6         |
| <b>Q.3 Solve Any Two of the following.</b>                                                                 |            | <b>12</b> |
| A) Explain the sequencing of pneumatic cylinders using cascade method.                                     | (CO3)      | 6         |
| B) Give the comparison between pneumatic systems and hydraulic systems.                                    | (CO3)      | 6         |
| C) Derive model equation of a mechanical system for mass and spring.                                       | (CO5)      | 6         |
| <b>Q.4 Solve Any Two of the following.</b>                                                                 |            | <b>12</b> |
| A) Explain with a block diagram of 8085 microprocessor and its architecture.                               | (CO4)      | 6         |
| B) Describe with a neat sketch Programmable Logic Controller (PLC).                                        | (CO4)      | 6         |
| C) Explain the 8081 micro controller with a neat sketch. Give a few applications of it.                    | (CO1)      | 6         |
| <b>Q.5 Solve Any Two of the following.</b>                                                                 |            | <b>12</b> |
| A) Explain with a neat sketch of Proportional- Integral -Derivative (PID) Controller.                      | (CO6)      | 6         |
| B) Why a Closed Loop Control System is preferred? Explain it with a suitable example.                      | (CO5)      | 6         |
| C) What is the significance of a Proportional- Derivative (PD) Controller? Explain it with an application. | (CO6)      | 6         |

\*\*\* End \*\*\*